PRODUCT SPACES AND RANDOM PROCESSES

Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be measurable spaces.

$$\underbrace{A \times B := \{(x,y) : x \in A,\ y \in B\}}_{\text{product of } A \text{ and } B}$$

If $A \in \mathcal{E}$ and $B \in \mathcal{F}$ then $A \times B$ is called a measurable rectangle.

Then $\mathcal{E} \otimes \mathcal{F} := \sigma(A \times B : A \times B \text{ measurable rectangles})$

The measurable space $(E \times F, \mathcal{E} \otimes \mathcal{F})$ is called the product of $(E, \mathcal{E})$ and $(F, \mathcal{F})$.

Lemma. It is true that

$$\mathcal{E} \otimes \mathcal{F} = \sigma(\hat{\mathcal{E}} \cup \hat{\mathcal{F}})$$

where

$$\hat{\mathcal{E}} = \{A \times F : A \in \mathcal{E}\}, \quad \hat{\mathcal{F}} = \{E \times B : B \in \mathcal{F}\}$$

The infinite case:

Let T be an arbitrary set (playing the role of index set). For each $t \in T$, let $(E_t, \mathcal{E}_t)$ be a measurable space. Let $x_t \in E_t$, for each $t \in T$. Then $(x_t)_{t \in T}$ (the collection) can be viewed as a function on T, especially if $(E_t, \mathcal{E}_t) = (E, \mathcal{E})$ for all $t \in T$, because $t \mapsto x_t$ can be seen as

a mapping
$$T \ni t \mapsto x_t \in E$$
The set F of all such functions, $x = (x_t)_{t\in T}$ is called the product space defined by $\{E_t, t \in T\}$. The notation, often, is
$$F =: \mathop{\times}_{t\in T} E_t$$
[T is a finite number of "positions"]

A rectangle in F is a subset
$$\mathop{\times}_{t\in T} A_t = \{x \in F : x_t \in A_t \text{ for each } t \in T\}$$
where A_t differs from E_t only for a finite number of t. The rectangle is measurable if $A_t \in \mathcal{E}_t$ for every t (for which A_t differs from E_t). Then σ(measurable rectangles) is called the product σ-Algebra and is denoted by $\bigotimes_{t\in T} \mathcal{E}_t$.

The resulting product space is
$$\left(\mathop{\times}_{t\in T} E_t, \bigotimes_{t\in T} \mathcal{E}_t\right).$$
In the case $(E_t, \mathcal{E}_t) = (E, \mathcal{E})$ for each $t \in T$ we use the notations
$$(E, \mathcal{E})^T = (E^T, \mathcal{E}^T).$$
Proposition. Let $(\Omega, \mathcal{H})$ be a measurable space. Let $(F, \mathcal{F}) = \bigotimes_{t\in T}(E_t, \mathcal{E}_t)$. For each $t \in T$, let f_t be a mapping from Ω to E_t. For each $\omega \in \Omega$, define $f(\omega)$ to be the point $(f_t(\omega))_{t\in T}$ in F.

Then, the mapping $f: \Omega \mapsto F$ is measurable relative to $\mathcal{A}$ and $\mathcal{F}$ iff f_t is measurable relative to $\mathcal{A}$ and $\mathcal{E}_t$ for every $t \in T$. $\square$

This justifies the definition:

Definition. Let $(\Omega, \mathcal{A}, P)$ be a probability space. A d-dimensional stochastic process indexed by $I \subset [0,\infty)$ is a family of r.v.'s $X_t : \Omega \mapsto \mathbb{R}^d$, $t \in I$. We write $X = (X_t)_{t \in I}$, I is called the index set, $\mathbb{R}^d$ the state space.

The only requirement of previous definition is that the functions X_t, $t \in I$ are measurable relatively to $\mathcal{A}$ and $\mathcal{B}(\mathbb{R}^d)$. An object so defined is too general to be usefull, especially in applications. We need more information on

$$(t,\omega) \mapsto X_t(\omega)$$

as a function of two variables. This is the reason why in the theory of probability particular classes of random processes are considered, e.g.,

- Diffusion processes
- Gaussian processes
- Markov processes
- Lévy processes
- Feller processes
- Martingales
- Semi-Markov processes
- Point-processes
- .....

Brownian motion belong to all these categories.

What is Brownian motion?

If you observe plant pollen in a drop of water through a microscope, you will see an incessant, irregular movement of the particles. This was observed by Gleichen-Russwurm to which the botanist Robert Brown refers when he wrote the first scientific papers on this. Brown pointed out the main features of the erratic motion of particles, qualitatively.

Brownian motion emerged in Physics, especially with Einstein. In mathematics we have contributions from

- Bachelier (1900), not fully rigorous
- Wiener (1923)
- Kolmogorov
- Lévy
- Donsker
- ....

We begin with a heuristic mathematical model for the erratic motion of particles (let us stay in 1-dim).
Assume that each particle
- starts at the origin
- changes position only at discrete times $k\Delta t$ where $\Delta t > 0$ is fixed and $k = 1, 2, \dots$
- moves Δx units to the left or to the right with the same probability

Letting Δt and Δx to zero in the appropriate way should give a random motion on the line $\mathbb{R}$ which is continuous.

Let us denote by X_t the random position of a particle at time $t \in [0,T]$. During the time $[0,T]$ the particle has changed its position $N := \lfloor T/\Delta t \rfloor$ times. The decision to move left or right can be modelled by a sequence of Bernoulli r.v.'s ε_k, $k=1,2\dots$ with

$$P(\varepsilon_k = 1) = 1/2 = P(\varepsilon_k = 0).$$

Hence, define

$S_N = \varepsilon_1 + \dots \varepsilon_N \longrightarrow$ number of right moves,

$N - S_N \longrightarrow$ number of left moves,

so that

$$\begin{aligned} X_T &= S_N \Delta x - (N - S_N)\Delta x \\ &= 2 S_N \Delta x - N \Delta x \qquad \circledast \\ &= \sum_{k=1}^{N} (2\varepsilon_k - 1)\Delta x \end{aligned}$$

is the position at time $T = N\Delta t$. Instead, for any couple of times $t = n\Delta t$, $T = N\Delta t$, we have

$$X_T = X_T - X_t + X_t - \underbrace{X_0}_{=0} =$$

$$= \sum_{n=1}^{N}{}' (2\varepsilon_n - 1)\Delta x - \sum_{n=1}^{m}{}' (2\varepsilon_n - 1)\Delta x + \sum_{n=1}^{m}{}' (2\varepsilon_n - 1)\Delta x$$

$$= \underbrace{\sum_{n=m+1}^{N}{}' (2\varepsilon_n - 1)\Delta x}_{X_T - X_t} + \underbrace{\sum_{n=1}^{m}{}' (2\varepsilon_n - 1)\Delta x}_{X_t - X_0}$$

Since the ε_n are iid the two increments $X_T - X_t$ and $X_t - X_0$ are indep. and further

$$X_T - X_t \sim X_{T-t} - X_0 \quad \text{since}$$

$$X_{T-t} = \sum_{n=1}^{N-m} (2\varepsilon_n - 1)\Delta x \sim \sum_{n=m+1}^{N}{}' (2\varepsilon_n - 1)\Delta x$$

sum of the same number of iid r.v.'s.

Denote $\sigma^2(t) := \operatorname{Var} X_t$. We have

$$\begin{aligned}
\sigma^2(T) &= \operatorname{Var}(X_T - X_t) + \operatorname{Var}(X_t - X_0) && \text{indep.} \\
&= \operatorname{Var}(X_{T-t} - X_0) + \operatorname{Var}(X_t) && \text{id. distr.} \\
&= \operatorname{Var}(X_{T-t}) + \operatorname{Var} X_t && X_0 = 0 \\
&= \sigma^2(T-t) + \sigma^2(t)
\end{aligned}$$

so that $\sigma^2(t)$ is linear, i.e.,

$$\sigma^2(T) = \sigma^2 T$$

where $\sigma > 0$ is called the diffusion coefficient.
However we can also directly compute

$$\operatorname{Var} X_T = \operatorname{Var} \sum_{k=1}^{N} (2\varepsilon_k - 1)\,\Delta x$$

$$= \sum_{k=1}^{N} \operatorname{Var}(2\varepsilon_k - 1)\,\Delta x \qquad \text{indep. of } \varepsilon_k$$

$$= \sum_{k=1}^{N} \operatorname{Var} 2\varepsilon_k \,\Delta x$$

$$= \sum_{k=1}^{N} 4(\Delta x)^2 \operatorname{Var} \varepsilon_k$$

$$= \cancel{4} N (\Delta x)^2 \cancel{1/4} \qquad \operatorname{Var} \varepsilon_k = 1/4$$

$$= \frac{T}{\Delta t} (\Delta x)^2 \qquad T = N \Delta t$$

$$= \sigma^2 T$$

and thus

$$\sigma^2 \cancel{T} = \cancel{T}/\Delta t \,(\Delta x)^2$$

i.e.,

$$\sigma^2 = \frac{(\Delta x)^2}{\Delta t}.$$

Using the computation above ⊛ we have

$$X_T = (2S_N - N)\,\Delta x = \frac{2S_N - N}{\sqrt{N}} \Delta x \sqrt{N}$$

$$= \frac{S_N - N/2}{\frac{1}{2}\sqrt{N}} \Delta x \sqrt{N} = \underbrace{\frac{S_N - E S_N}{\sqrt{\operatorname{Var} S_N}}}_{=: S_N^*} \Delta x \sqrt{N}$$

$\Delta x = \sqrt{\Delta t}\,\sigma$

$$= S_N^* \sqrt{N \Delta t}\,\sigma$$

$$= S_N^* \sigma \sqrt{T}$$

A simple application of CLT then yields to

$$X_T = \sqrt{T}\,\sigma\, S_N^* \xrightarrow{N\to+\infty} \sqrt{T}\,\sigma\, G$$

where $G \sim N(0,1)$. Note that $N \to +\infty$ means $\Delta x \to 0$, $\Delta t \to 0$, $\frac{\Delta x}{\sqrt{\Delta t}} = \sigma$ constant.

It follows that the particle position

$$\lim_{\Delta x, \Delta t \to 0} X_T = B_T$$

is $N(0, T\sigma^2)$, while for each t, $B_t \sim N(0, t\sigma^2)$.

This motivates the following.

Definition. A d-dimensional Brownian motion $B = (B_t)_{t\geq 0}$ is a stochastic process indexed by $[0,\infty)$ taking values in $\mathbb{R}^d$ s.t.

0) $B_0(\omega) = 0$ for a.e. ω

1) $B_{t_n} - B_{t_{n-1}}, \dots, B_{t_1} - B_{t_0}$ are indep. for any $n \geq 0$, $0 = t_0 < t_1 < t_2 < \dots < t_n < +\infty$.

2) $B_t - B_s \sim B_{t+h} - B_{s+h}$ for all $0 \leq s < t$, $h \geq -s$

3) $B_t - B_s \sim N(0, (t-s)\mathbb{1})$

4) $t \mapsto B_t(\omega)$ is continuous for all ω.

We will also speak of a BM if the index set is an interval of the form $[0,T]$.
We say that $(x+B_t)_{t\geq 0}$, $x\in\mathbb{R}^d$, is a d-dimensional BM started at x.
As we will see the definition is redundant, i.e.,

0) - 3) $\Rightarrow$ 4) for a.a. ω

0) - 2) and 4) $\Rightarrow$ 3)

Now we prove the equivalence of a different formulation of the notion "independent increments" which pass through the use of generated σ-algebras and will be useful later.
Then we should address the problem of finding a process fitting the definition, but we leave this for later. Before we study some properties of BM which follows from the definition and have a direct relation with the fact that BM is a Gaussian process.

Here is an equivalent formulation of the notion of "indep. increments".

Theorem. Let $(X_t)_{t\geq 0}$ be a stochastic process in $\mathbb{R}^d$. Then

X_t has indep. increments $\Longleftrightarrow$ $X_t - X_s$ is indep from $\sigma(X_u, u \leq s)$ for any $s \geq 0$

Recall that

$$\sigma(X_u, u \leq s) := \sigma\Big(\bigcup_{u \leq s} \sigma(X_u)\Big)$$

Proof. We use Kac's theorem, i.e., for $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$E e^{i\langle \xi, X\rangle} = E e^{i\xi_1 X_1} E e^{i\xi_2 X_2} \Longleftrightarrow x_1, x_2 \text{ independent}$$

Proof of "$\Leftarrow$":

$$E e^{i\sum_{k=1}^{m} \langle \xi_k, X_{t_k} - X_{t_{k-1}}\rangle} = E\Big[E\Big[\exp\Big(i\sum_{k=1}^{m} \langle \xi_k, X_{t_k} - X_{t_{k-1}}\rangle\Big)\Big|\mathcal{F}^X_{t_{m-1}}\Big]\Big]$$

tower property

$\sigma(X_u, u \leq t_{m-1})$

$X_{t_k} - X_{t_{k-1}}$ is $\mathcal{F}_{t_{m-1}}$-measurable for $k \leq m-1$

$$= E\Big[\exp\Big(i\sum_{k=1}^{m-1} \langle \xi_k, X_{t_k} - X_{t_{k-1}}\rangle\Big) E\Big[\exp\big(i\langle \xi_m, X_{t_m} - X_{t_{m-1}}\rangle\big)\Big|\mathcal{F}^X_{t_{m-1}}\Big]\Big]$$

$$= E\Big[\exp\Big(i\sum_{k=1}^{m-1} \langle \xi_k, X_{t_k} - X_{t_{k-1}}\rangle\Big)\Big] E\Big[\exp\big(i\langle \xi_m, X_{t_m} - X_{t_{m-1}}\rangle\big)\Big]$$

$X_{t_m} - X_{t_{m-1}}$ indep of $\mathcal{F}^X_{t_{m-1}}$ by hypotheses.

$= \quad \dots\dots \quad$ iterating $\quad \dots\dots$

$$= \prod_{k=1}^{n} E \exp\left(i \langle \xi_k, X_{t_k} - X_{t_{k-1}} \rangle \right).$$

Hence the implication follows by Kac's theorem.

Proof of "$\Rightarrow$" [NB: $\bigcup_{n \in \mathbb{N}} \sigma(X_n)$ is not a σ-system]

LEMMA: We ~~have~~ that

$$\mathcal{G}_s := \bigcup_{n} \bigcup_{0 \le s_1 \le \dots \le s_n \le s} \sigma(X_{s_1} \dots X_{s_n})$$

$\rightarrow$ s can be $+\infty$

is a π-system which generates $\mathcal{F}_s^X$.

Therefore it is sufficient to check independence on the π-system, i.e., to check that

$\sigma(X_t - X_s)$ is indep from $\mathcal{G}_s$

or equivalently

$X_t - X_s$ is indep from $(X_{s_1}, \dots, X_{s_n})$, for any $n \in \mathbb{N}$ and $0 \le s_1 \le \dots \le s_n \le s$.

First we prove that $\mathcal{G}_s$ is a π-system.

If $A \in \mathcal{G}_s$ and $B \in \mathcal{G}_s$ there exist $m, n \in \mathbb{N}$ and $0 \le s_1 \le \dots \le s_m \le s$ and $0 \le t_1 \le \dots \le t_n \le s$ such that $A \in \sigma(X_{s_1} \dots X_{s_m})$ $B \in \sigma(X_{t_1} \dots X_{t_n})$.

Let $0 \le u_1 \le \dots \le u_N \le s$ be an increasing enumeration of the refinement $\{s_j, j \le m\} \cup \{t_j, j \le n\}$ so that $A, B \in \sigma(X_{u_1} \dots X_{u_N})$ and thus

$$A \cap B \in \sigma(X_{u_1} \dots X_{u_N}) \subset \mathcal{G}_s$$

This proves that $\mathcal{G}_s$ is a π-system. Now we prove that it generates $\mathcal{F}_s^X$.

It is clear that $(X_{s_1} \dots X_{s_n})$ is $\mathcal{F}_s^X$-measurable and so $\mathcal{G}_s \subset \mathcal{F}_s^X$ which implies that $\sigma(\mathcal{G}_s) \subset \mathcal{F}_s^X$. Furthermore, by definition of $\mathcal{G}_t$, one has $\sigma(X_t) \subset \mathcal{G}_s$ for each $t \in [0,s]$ so that X_t is $\mathcal{G}_s$-measurable for each $t \in [0,s]$.

Since $\mathcal{F}_s^X$ is the smallest σ-algebra which makes all the $X_t, t \le s$ measurable we find that $\mathcal{F}_s^X \subset \sigma(\mathcal{G}_s)$.

We conclude that $\mathcal{F}_t^X = \sigma(\mathcal{G}_t)$.

We are now ready to prove that $X_t - X_s$ and $(X_{s_1} \dots X_{s_n})$ are independent.

By assumptions the r.v.'s $X_{s_1} - X_{s_0}, \dots (X_{s_n} - X_{s_{n-1}}),$ $X_t - X_s$ are independent. Since

$$X_{s_k} = \sum_{j=1}^{k} (X_{s_j} - X_{s_{j-1}}), \qquad k = 1 \dots n,$$

we see that $(X_t - X_s)$ indep of $(X_{s_1} \dots X_{s_n})$.